

Exploratory Three-Bat MOI and Swing-Speed Pilot

A transparent record of a homemade physical-pendulum measurement and a small tee-swing session. The note preserves the data while separating it from the stronger evidence used in the main bat-fitting article.

Scope: one hitter, three non-matched bats, estimated MOI, 25 tee swings, one commercial inertial sensor, and no simultaneous exit-velocity or impact-location measurement. This is an exploratory methods note - not a validation study and not evidence that any tested bat is optimal.

Why preserve an imprecise pilot?

A weak experiment can still be useful when its boundaries are explicit. This note records the apparatus, assumptions, raw sensor outputs, and descriptive calculation so the work can be inspected or improved without asking the main article to carry more evidentiary weight than the study deserves.

What it can show

Across these three bat averages, the higher estimated-MOI bats were associated with lower measured bat speed. That direction is consistent with prior research.

What it cannot show

It cannot identify the best bat, estimate an exit-velocity optimum, measure barrel control, isolate MOI from length or construction, validate the sensor, or generalize to another hitter.

Jacob Chin | September 2026 | Exploratory technical note

1. Equipment and Methods

Bat measurements

Each bat's length, total mass, and balance point were measured. The bat was then suspended as a physical pendulum from a pivot located six inches from the knob. The mean oscillation period was used to estimate MOI about that pivot. All three bats were measured with the same homemade setup.

Small-angle physical-pendulum relation

$I_p = m g d T^2 / (4 \pi^2)$

where I_p is MOI about the pivot, m is bat mass, d is the pivot-to-center-of-mass distance, and T is the oscillation period.

Bat	Length	Mass	Balance point	Mean period	Estimated MOI
DeMarini Voodoo One (2022)	31.9 in	867 g	18.7 in	1.62 s	9,970 oz-in^2
Rawlings 5150 (2019)	32.0 in	856 g	19.9 in	1.70 s	11,870 oz-in^2
Marucci AP5 wood bat	33.1 in	918 g	21.8 in	1.68 s	14,120 oz-in^2

The absolute MOI values are estimates. Pivot friction, timing resolution, release amplitude, alignment, and balance-point measurement were not controlled to laboratory standards. The values should not be treated as certified specifications for these bat models.

Swing protocol

One hitter took maximum-intent swings from a stationary tee placed at approximately mid-thigh height over the middle of the plate. Bats were rotated in a 2-2-2 sequence to reduce simple order and fatigue effects. Eight swings were recorded for the AP5, eight for the 5150, and nine for the Voodoo.

A Blast Motion sensor reported bat speed, rotational acceleration, time to contact, and power. The AP5 was entered in the application as 33 inches; the Voodoo and 5150 were entered as 32 inches. No launch monitor, optical tracking, direct impact-location system, or ball-flight measurement was synchronized with the sensor.

Analysis

Arithmetic means were calculated for each reported metric. The three bat-speed means were fitted on a log-log scale to $v = C I^{-n}$. Because the regression contains only three aggregated observations, n and R^2 are reported descriptively, not as inferential statistics.

2. Descriptive Results

0.224	0.961	25
Fitted n	Three-point R^2	Recorded swings

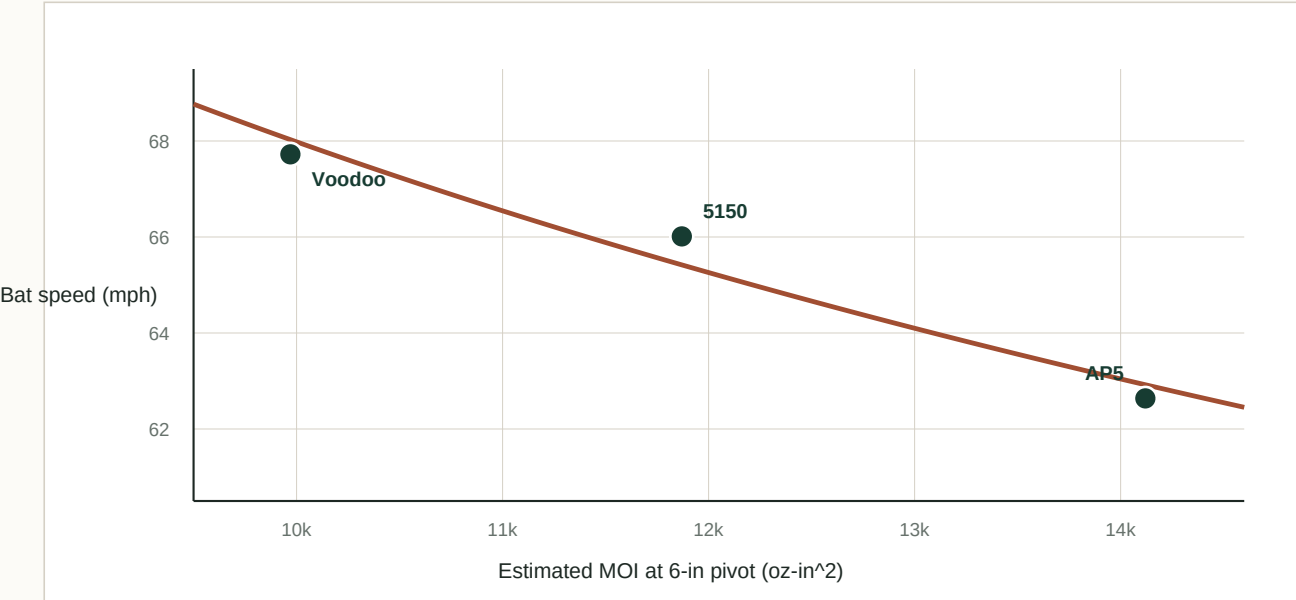


Figure 1. Mean sensor-reported bat speed versus estimated MOI. The fitted line is a compact description of these three means only. Different bat lengths, materials, constructions, and measurement errors are confounded with MOI.

Mean sensor outputs

Bat	Swings	Bat speed (mph)	Rot. accel. (g)	Time to contact (s)	Power (kW)
AP5	8	62.64	15.38	0.149	3.32
5150	8	66.01	16.34	0.139	3.96
Voodoo	9	67.72	15.96	0.140	4.20

The time-to-contact entries were reported by the device only to hundredths of a second. The third decimal in the averages is mathematical output from averaging rounded values, not millisecond measurement precision.

Interpretation: estimated MOI and mean bat speed moved in the expected opposite directions, and the fitted n was similar in magnitude to published group estimates. The agreement is interesting but not validating: three confounded points can produce a visually strong fit by chance or shared measurement error.

3. Complete Sensor Output

All recorded swings are reproduced below in the order supplied. No swing-level observation was removed from the summary.

Bat	Swing	Bat speed	Rot. accel.	Time to contact	Power
AP5	1	63.5 mph	15.9 g	0.15 s	3.42 kW
AP5	2	63.3 mph	14.4 g	0.15 s	3.31 kW
AP5	3	62.4 mph	14.5 g	0.15 s	3.22 kW
AP5	4	62.6 mph	16.0 g	0.15 s	3.29 kW
AP5	5	62.4 mph	13.9 g	0.15 s	3.22 kW
AP5	6	63.5 mph	16.6 g	0.14 s	3.57 kW
AP5	7	61.8 mph	16.1 g	0.15 s	3.28 kW
AP5	8	61.6 mph	15.6 g	0.15 s	3.27 kW
5150	1	65.7 mph	16.4 g	0.14 s	3.88 kW
5150	2	67.5 mph	16.0 g	0.14 s	4.09 kW
5150	3	66.5 mph	17.4 g	0.14 s	4.03 kW
5150	4	65.9 mph	16.7 g	0.14 s	3.90 kW
5150	5	68.0 mph	14.5 g	0.14 s	4.27 kW
5150	6	65.3 mph	18.2 g	0.13 s	4.05 kW
5150	7	64.5 mph	14.9 g	0.14 s	3.74 kW
5150	8	64.7 mph	16.6 g	0.14 s	3.75 kW
Voodoo	1	67.4 mph	15.9 g	0.14 s	4.02 kW
Voodoo	2	69.2 mph	16.1 g	0.14 s	4.30 kW
Voodoo	3	69.2 mph	15.5 g	0.14 s	4.42 kW
Voodoo	4	68.0 mph	15.2 g	0.14 s	4.15 kW
Voodoo	5	67.3 mph	15.2 g	0.14 s	4.18 kW
Voodoo	6	68.6 mph	16.0 g	0.14 s	4.35 kW
Voodoo	7	65.5 mph	15.9 g	0.14 s	3.96 kW
Voodoo	8	67.3 mph	17.7 g	0.14 s	4.18 kW
Voodoo	9	67.0 mph	16.1 g	0.14 s	4.20 kW

Metric labels and units reproduce the device output supplied for this note. Rotational acceleration and power were not independently reconstructed from optical motion data.

4. Limitations and Appropriate Use

This note is most useful as a record of an early attempt and as a checklist for a better experiment. The following limitations are not minor qualifications; together they prevent the data from answering which bat was best.

1. One hitter and three bats cannot establish a general or individual optimum.
2. The bats differed in length, mass, material, construction, model, and mass distribution.
3. The homemade pendulum did not control pivot friction, alignment, amplitude, or timing with laboratory precision.
4. The sensor was not synchronized with optical motion capture, a launch monitor, or an impact-location system.
5. Stationary tee swings remove pitch perception, timing uncertainty, and most of the adjustability problem relevant to barrel control.
6. Exit velocity, launch angle, spray direction, contact efficiency, and impact location were not measured.
7. The power-law fit uses only three aggregated points and should not be extrapolated outside the observed range.
8. The experiment does not isolate whether the observed speed differences came from MOI, length, construction, familiarity, or measurement error.

Relationship to the main article

The main article, *How Do You Know Which Bat Is Best for You?*, relies on published research to develop a bat-fitting framework around exit velocity and barrel control. This pilot is not evidence for that article's hypothesis. It is optional background showing why future work needs matched bats, precise MOI measurement, optical tracking, direct exit velocity, impact location, and randomized pitch conditions.

References

1. Smith, L., Broker, J., and Nathan, A. (2003). *A Study of Softball Player Swing Speed*. baseball.physics.illinois.edu/SwingSpeed.pdf
2. Russell, D. A. *Moment-of-Inertia of Baseball and Softball Bats*. acs.psu.edu/drussell/bats/bat-moi-details.html
3. Nagami, T. et al. (2022). *Accuracy and Error Trends of Commercially Available Bat Swing Sensors in Baseball*. pmc.ncbi.nlm.nih.gov/articles/PMC8879135/

Bottom line: the pilot reproduced the expected direction of the MOI-speed relationship, but its measurement quality and design do not support a bat-fitting conclusion. Preserve it as an exploratory note; build the scientific argument from controlled published research and a better future study.